

Jet Engine Hospital

A Leakage-Safe Early-Warning System for NASA C-MAPSS

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Executive summary

Jet Engine Hospital is a multi-task predictive-maintenance system for NASA C-MAPSS turbofan engines. At every observed cycle it estimates Remaining Useful Life (RUL), produces calibrated probabilities of failure within 10, 20, and 30 cycles, measures abnormality without fitting anomaly detectors on failure labels, and combines those signals into a traceable **CONTINUE / INSPECT / STOP** recommendation. FD001 is the mandatory foundation and FD003 is the Stage 2 multi-fault challenge. Both use the same engine-level split, causal feature pipeline, validation roles, metrics, uncertainty methods, and decision policy. FD004 is included as a bonus combined-condition and multi-fault experiment.

The selected window Gradient Boosting RUL model achieves official-test RMSE 7.76 on FD001 and 6.57 on FD003. Its nominal 90% split-conformal intervals cover 94.0% and 95.6% of capped targets, respectively. At the 30-cycle horizon, calibrated PR-AUC is 0.930 on FD001 and 0.936 on FD003. Isolation Forest is validation-selected for anomaly detection on both subsets, but its high miss rates demonstrate why anomaly evidence is used for inspection priority rather than as a failure probability. The deployed Gradio interface exposes the full evidence chain, uncertainty, model disagreement, and the exact rule that produced the recommendation.

The central engineering objective is defensibility rather than a single headline metric. Engines are separated before preprocessing or window creation; transformations are fitted on training engines only; calibration and operational thresholds use held-out validation engines; official test data are evaluated only after the pipeline is frozen.

1. Problem formulation and data audit

Each C-MAPSS row describes one engine at one cycle using an engine identifier, cycle, three operating settings, and 21 sensor readings. Files are whitespace-separated and have no header, so the loader assigns all 26 names explicitly. It rejects unknown subsets, incorrect column counts, missing or non-finite values, duplicate (engine_id, cycle) keys, and discontinuous cycle sequences.

For a training trajectory, the failure cycle is its final observed cycle T_i and the uncapped target is $RUL_{i,t} = T_i - t$. Test trajectories stop before failure; the official RUL_FD00x.txt value is added to the final observed cycle to construct a virtual failure cycle. Counts and official engine order are asserted before labels are attached. Three distinct classification labels are then created: $I(RUL \leq 10)$, $I(RUL \leq 20)$, and $I(RUL \leq 30)$.

FD001 contains 100 training and 100 official-test engines under one condition and one fault mode. FD003 contains 100 training and 100 test engines under one condition and two fault modes. The supplied FD004 files contain 249 training and 248 test engines, the reverse of the counts printed in the project table; 248 official RUL values confirm the file-derived test count. This discrepancy is documented rather than bypassed.

The notebook reports row and engine counts, sequence-length distributions, missing and non-finite values, duplicate keys, continuity, train-only variances, near-constant sensors, operating-setting distributions, class balance for every horizon, and a table of filtering decisions with affected rows and engines. Several sensor trajectories are plotted to show scale, noise, and degradation direction. No engines or rows are silently removed.

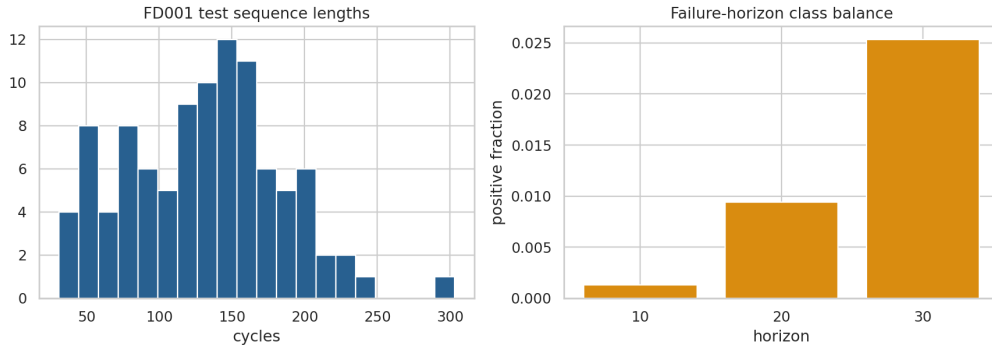


Figure 1: FD001 data audit and sequence diagnostics

2. Leakage controls and causal features

Official-training engine IDs are divided into 70% training and 30% validation using seed 42. Validation engines are divided again into `val_tune` and `val_calib`. Training fits preprocessing and estimators; `val_tune` selects caps, feature representations, model families, and anomaly families; `val_calib` fits conformal quantiles, Platt calibrators, probability thresholds, anomaly thresholds, and persistence rules. Official-test engines remain separate. Persisted identities include source roles, such as `FD001:official_test:1`, because train and test files reuse local engine numbers.

Feature engineering is performed independently within each engine and uses only rows at or before the prediction cycle. The current-cycle baseline is compared with trailing windows containing current value, mean, standard deviation, minimum, maximum, ordinary least-squares slope, first difference, and exponentially weighted mean. Windows are right-aligned and never cross an engine or split boundary. Scaling, sensor selection, and optional condition processing are fitted on training engines only.

A direct causality test perturbs every row after a chosen cycle and verifies that all features at and before that cycle remain unchanged. Additional checks cover shuffled input alignment, engine-boundary isolation, duplicate keys, discontinuities, NaN/Inf rejection, and train-only scaler state. The rolling-slope implementation uses vectorized rolling sums rather than a Python callback per row; this preserves the causal formula while substantially reducing execution time.

Two required ablation dimensions are present. Window lengths are compared to the current-cycle representation, and operating settings are included or removed under a fixed model family. FD004 additionally compares global scaling with training-only condition-aware normalization.

3. RUL regression and uncertainty

RUL candidates include Ridge, degree-two Polynomial Ridge, Random Forest, and Gradient Boosting regressors under identical engine roles. Both current-cycle and explicit-window representations are evaluated. RUL caps are selected using validation engines; models are never selected or refitted using official-test outcomes. Comparisons record MAE, RMSE, R^2 , the NASA/PHM asymmetric score, fit time, serialized size, and per-row inference latency.

Subset	Selected model	MAE	RMSE	R^2	Mean PHM	90% PI coverage	Width
FD001	Window Gradient Boosting	4.89	7.76	0.831	0.929	0.940	21.70
FD003	Window Gradient Boosting	3.51	6.57	0.841	0.715	0.956	20.36

The PHM sign convention is prediction minus truth. A positive error overestimates safe life and receives the stronger late-maintenance penalty. The notebook verifies this with a worked +10 versus −10 cycle example. Results are also split into early, middle, and near-failure regions; mid-life is hardest because the model must locate the departure from the capped healthy plateau.

Uncertainty uses split conformal prediction. Absolute residuals on val_calib determine the finite-sample adjusted 90th percentile q . The interval is $[\hat{y} - q, \hat{y} + q]$, clipped to the selected target support. Coverage and average width are reported overall, by engine, and by RUL region. Confidence intervals for primary metrics use 1,000 engine-level bootstrap draws so dependent cycles from the same engine are never resampled as independent rows.

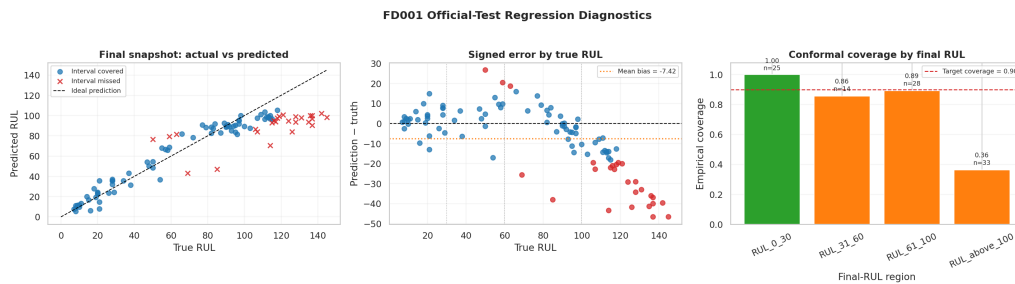


Figure 2: FD001 official-test RUL diagnostics

4. Failure-horizon classification

Separate models are trained for 10-, 20-, and 30-cycle failure horizons. Logistic Regression is the transparent baseline and Random Forest is the nonlinear comparison. Selection uses `val_tune` PR-AUC. Each selected estimator receives its own Platt calibrator fitted on `val_calib`; a cumulative maximum then enforces the logical relation $P_{10} \leq P_{20} \leq P_{30}$. Decision thresholds minimize a declared validation cost that weights false negatives ten times more than false positives, rather than assuming 0.50.

Subset	Horizon	PR-AUC	Precision	Recall	F1	Brier
FD001	10	0.765	0.246	1.000	0.395	0.00071
FD001	20	0.860	0.488	0.959	0.647	0.00279
FD001	30	0.930	0.535	0.991	0.695	0.00585
FD003	10	0.662	0.419	1.000	0.590	0.00065
FD003	20	0.932	0.539	0.967	0.692	0.00170
FD003	30	0.936	0.658	0.979	0.787	0.00393

The notebook exports precision-recall curves, reliability diagrams, confusion matrices, threshold diagnostics, before/after Brier scores, per-engine distributions, and 95% engine-bootstrap intervals for each horizon. Longer horizons generally warn earlier and provide more positive examples, but can increase ambiguity and false alarms. The 10-cycle label is rare in truncated test histories, so its precision and bootstrap stability require particular caution.

FD003 Official-Test Classification Diagnostics

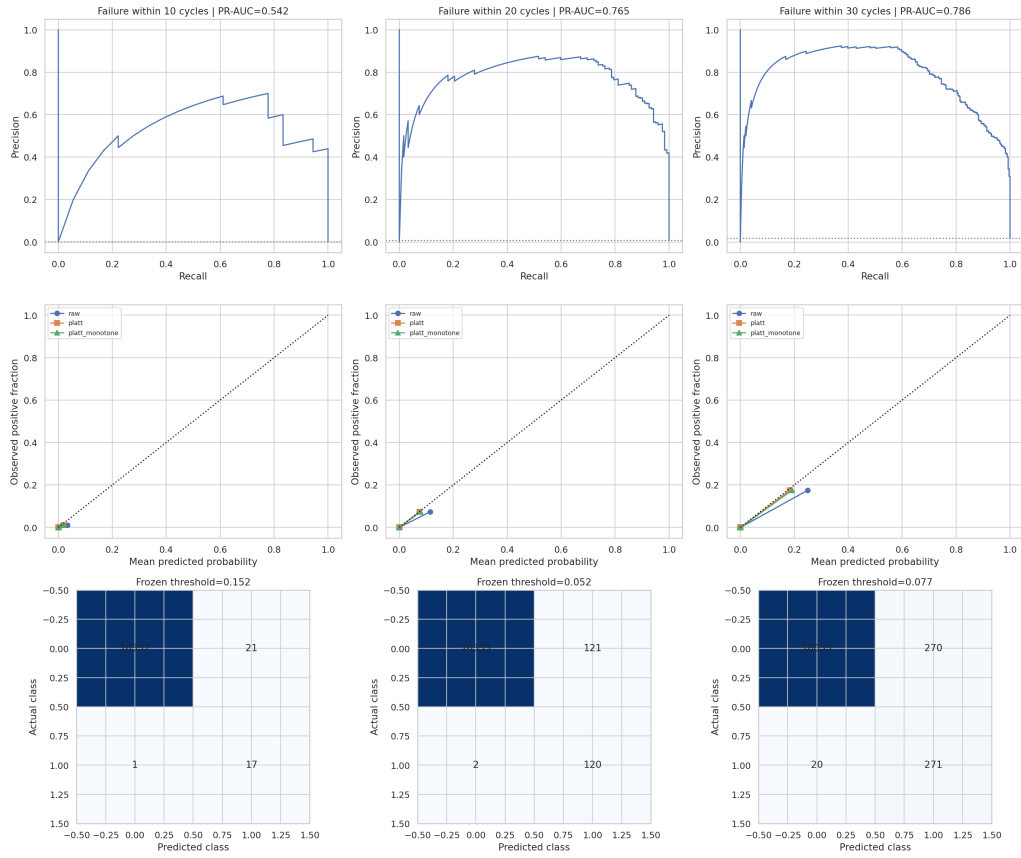


Figure 3: FD003 classification diagnostics

5. Unsupervised anomaly detection and warning cost

Four required detectors share the scaled causal feature space: Isolation Forest, Local Outlier Factor with `novelty=True`, RBF One-Class SVM, and PCA reconstruction error. They are fitted only on the first 30 observed cycles of training engines. This healthy proxy uses cycle position, not RUL or a failure-horizon label. One-Class SVM uses a deterministic, engine-balanced cap for tractable runtime.

All score directions are oriented so larger means more abnormal. Because detector scales are incomparable, raw scores are mapped through the frozen healthy-reference empirical CDF. The resulting normalized score is a percentile in $[0, 1]$, not a probability. Validation compares detector-family proxy PR-AUC and threshold sensitivity. The selected operational rule also requires an m -of- n persistent crossing to prevent a single sensor spike from causing an action.

For an engine failing at T_i and first persistently alerting at τ_i , lead time is $L_i = T_i - \tau_i$. The reported cost is

$$C = 100I(\text{miss}) + 5 \max(0, h - L_i) + \max(0, L_i - h).$$

Miss rate, late delay, early burden, and total cost remain separate. Isolation Forest is selected for FD001 and FD003, but official-test persistent-alert miss rates are 0.72 and 0.64. This is an important negative result: unsupervised evidence is useful for inspection prioritization, but it is not sufficiently reliable to trigger STOP alone. PCA contributions are still exported to indicate which feature groups drive reconstruction error.

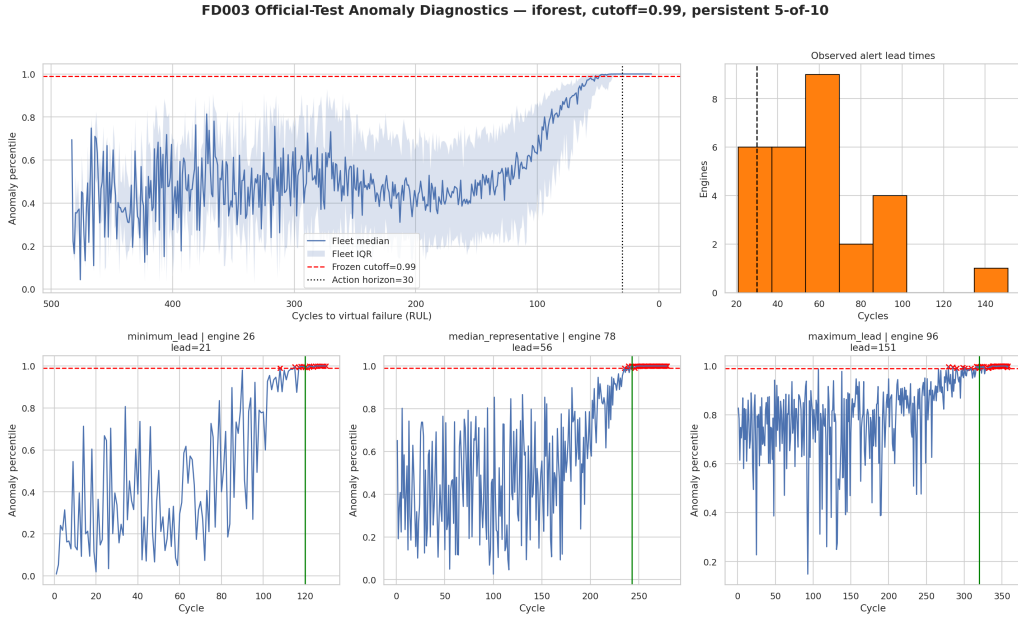


Figure 4: FD003 anomaly trajectories and warning diagnostics

6. Integrated maintenance policy and dashboard

Each evidence family first produces an auditable action. RUL recommends STOP when the lower conformal bound is at most 10 cycles and INSPECT when it is at most 30. Classification recommends STOP when calibrated 10-cycle risk crosses its validation threshold and INSPECT when either 20- or 30-cycle risk crosses. Anomaly detection can recommend INSPECT only after its persistent rule fires. The final action uses severity precedence STOP > INSPECT > CONTINUE.

Confidence is deliberately not presented as another probability. It is HIGH when all three evidence-family actions agree with the final action, MODERATE for two of three, and LOW when a single higher-severity family determines the result. The dashboard shows the family actions and agreement count, so disagreement is visible rather than silently collapsed. Every row also stores the exact trigger explanation.

The Gradio application lets the operator select subset, engine, cycle, and sensor. It shows the sensor and RUL histories with the current cycle highlighted; RUL point estimate and interval; calibrated 10/20/30-cycle risks and thresholds; normalized anomaly score, healthy-reference percentile, threshold margin, and persistence; final action, confidence, and trigger; and dataset, version, feature window, and training timestamp. Expensive inference is precomputed, keeping startup memory and latency modest.

Deployment parity is checked independently of the interface. Frozen systems are reloaded, official-test features and predictions are recomputed, and numeric outputs, family actions, confidence, and recommendations are compared with every exported dashboard row. The app does not train or substitute a separate model.

7. Stage 2 and FD004 bonus

FD003 is the required Stage 2 multi-fault challenge. It reuses the FD001 split policy, feature definitions, model families, calibration, uncertainty, persistence search, metrics, bootstrap, and dashboard contract. Supervised performance remains competitive: RUL RMSE improves from 7.76 to 6.57 and 30-cycle PR-AUC changes from 0.930 to 0.936. The two fault mechanisms nevertheless produce heterogeneous terminal signatures and a different anomaly-warning burden. Because NASA supplies no per-engine physical fault label, train-derived latent groups are reported only as descriptive behavior proxies and are never used as predictive inputs.

FD004 is a bonus experiment with six conditions and two fault modes. A train-only model clusters operating settings into six regimes and normalizes retained sensors within the assigned regime. Validation compares global and condition-aware representations under fixed models, then compares nonlinear condition-aware candidates. The selected condition-aware model obtains official-test MAE 5.71, RMSE 10.64, R^2 0.618, and 90% interval coverage 0.957 with average width 32.62 cycles.

Transfer experiments train Ridge/window-15 systems on FD001, FD002, and FD003 and apply them to the same FD004 target without target-domain refitting. Single-condition FD001 and FD003 transfer fails severely because context shift is interpreted as degradation. FD002 transfer is better because its operating regimes match, but it still lacks the second fault mechanism and remains worse than dedicated FD004 training. Regime-level diagnostics support one global validation-selected anomaly threshold rather than post-test regime-specific tuning.

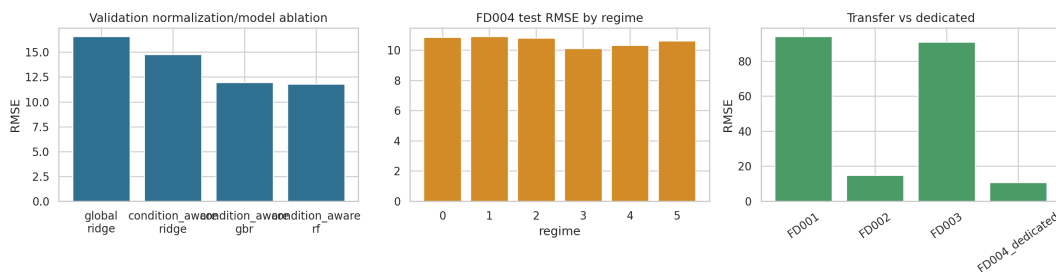


Figure 5: FD004 normalization, regime, and transfer comparisons

8. Limitations, reproducibility, and conclusion

C-MAPSS is simulated and does not establish airworthiness for real aircraft. Official test trajectories are right-truncated, so absence of an observed warning can reflect censoring rather than confirmed operational failure. RUL caps improve learnability but can underpredict genuinely long remaining life. Split-conformal intervals are marginal and pointwise, not simultaneous guarantees across a trajectory. Calibration partitions contain relatively few independent engines. FD003 and FD004 do not provide physical fault labels per engine, limiting fault-specific error analysis.

Reproducibility is enforced through fixed seeds, source-qualified engine splits, train-only transformations, explicit validation roles, schema checks, atomic contracts, artifact hashes, and notebook-to-app parity. The complete submission notebook executes from raw C-MAPSS inputs through final comparisons and deployment exports. Generated splits, processed contracts, models, figures, and app data are outputs rather than hidden inputs.

The project satisfies the requested evidence chain: data audit; causal window features; linear, polynomial, forest, and boosting regression baselines; separate calibrated failure horizons; four label-free anomaly detectors; conformal uncertainty; engine-bootstrap intervals; persistent warning analysis; asymmetric component costs; a traceable maintenance policy; FD003 Stage 2 comparison; and a deployment-ready Gradio dashboard. Its principal safety feature is not model complexity but the separation of training, selection, calibration, and final evaluation, together with visible uncertainty and disagreement.

References

1. A. Saxena, K. Goebel, D. Simon, and N. Eklund, “Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation,” *PHM 2008*.
2. A. Saxena and K. Goebel, “Turbofan Engine Degradation Simulation Data Set,” NASA Ames Prognostics Data Repository.
3. D. K. Frederick, J. A. DeCastro, and J. S. Litt, *User’s Guide for the Commercial Modular Aero-Propulsion System Simulation*, NASA/TM—2007-215026, 2007.